

Interim Report

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1 Introduction

1.1 Background

The global coffee industry remains a significant contributor to the economies of many developing countries, supporting millions of smallholder farmers worldwide. Coffee production involves several critical stages, from cultivation and harvesting to processing and packaging. Among these stages, the sorting of coffee berries is one of the most labor intensive and quality-sensitive processes. Traditionally, coffee berry sorting has relied heavily on manual visual inspection, where workers separate berries based on ripeness, size, and visible defects. This approach is time-consuming, inconsistent, and costly, often leading to variations in coffee quality and increased operational expenses. Manual sorting methods are highly susceptible to human error and fatigue, which can result in misclassification of berries and reduced overall product quality. Furthermore, the labor-intensive nature of manual sorting limits scalability and efficiency, particularly during peak harvest periods when labor shortages are common. These challenges highlight the need for automated sorting solutions capable of performing classification tasks with higher accuracy, consistency, and throughput. Recent advancements in computer vision, machine learning, and embedded systems have enabled the development of automated coffee berry sorting machines that utilize image-based detection and real-time classification techniques. Previous work by fifth-year mechatronic engineering students resulted in the development of an image-based coffee berry sorting machine that incorporates a vibrating conveyor system, optical detection, and air ejection mechanisms controlled by a Raspberry Pi-based processing unit. Although the system successfully demonstrated automated sorting, practical challenges were identified during real-time operation, particularly related to motion blur, processing latency, and limited computational throughput. The visual diversity in harvested coffee berries, including variations in color, size, and surface texture, as illustrated in Figure 1.1, presents a significant challenge for accurate automated classification. This project proposes the modification and improvement of the existing coffee berry sorting machine by addressing the identified limitations in image acquisition, processing speed, and real-time classification performance. By enhancing the vision processing pipeline, optimizing algorithms, and evaluating



Figure 1: Harvested coffee berry variation showing visual diversity in color, size, and surface texture

alternative hardware or processing architectures, the project aims to improve sorting accuracy and throughput while maintaining a cost-effective and scalable design suitable for small- and medium-scale coffee processing applications.

1.2 Problem Statement

The global coffee industry plays a vital role in the economic development of many coffee-producing countries, including Kenya, where coffee farming supports a large population of smallholder farmers. The quality of coffee berries at the processing stage directly determines the final product grade, market value, and competitiveness in both local and international markets. One of the most critical and labor-intensive stages in coffee processing is the sorting of coffee berries based on ripeness, color, size, and the presence of defects. Traditionally, coffee berry sorting has been carried out manually through visual inspection by human operators. While this method requires minimal capital investment, it is inherently inefficient, inconsistent, and highly dependent on human judgment, which varies due to fatigue, experience, and environmental conditions. As a result, manual sorting often leads to misclassification of berries, reduced product quality, and increased operational costs. These challenges become more pronounced during peak harvest seasons, when large volumes of berries must be processed within limited timeframes. In response to these limitations, recent final-year mechatronics projects have explored automated coffee berry sorting solutions using machine vision and robotic actuation. One such system employs a vibrating conveyor to ensure consistent berry flow, a chute for alignment, and an image-based detection and tracking system coupled with an air-ejection mechanism for sorting. The system operates under the control of a Raspberry Pi-based processing unit, leveraging computer vision and machine learning techniques to classify individual coffee berries as they move through the sorting process. Despite the successful demonstration of automated sorting, the existing system exhibits several critical technical limitations that hinder its overall performance and scalability. The natural appearance of harvested coffee berries presents significant visual variability in color, texture, size, and surface defects.

These variations, combined with the dynamic motion of berries on the conveyor, introduce challenges such as motion blur, inconsistent illumination, and overlapping objects, which negatively affect image clarity and classification accuracy. Furthermore, the reliance on real-time image processing using a Raspberry Pi introduces processing bottlenecks. The limited computational capacity of the platform results in delays during frame acquisition, preprocessing, classification, and decision execution. Although the previous implementation attempted to mitigate these issues through optimization techniques such as TensorFlow Lite and multithreading, the system still struggles to achieve higher throughput without compromising accuracy. This limitation directly affects sorting speed and reduces the system's suitability for larger processing volumes. The core problem, therefore, lies not in the mechanical handling of coffee berries but in the limitations of the vision system, processing architecture, and real-time control strategy used in the existing model. There is a clear need to improve the system's ability to accurately and consistently classify visually diverse coffee berries at higher speeds while maintaining affordability and practicality for small-scale agricultural applications. This project seeks to address these challenges through the modification and improvement of the existing automated coffee berry sorting system. By enhancing the image acquisition strategy, optimizing or upgrading the processing architecture, and refining the detection-to-actuation pipeline, the project aims to achieve higher throughput, reduced latency, and improved classification accuracy.

1.3 Objectives

Main Objectives

1. Improve an existing coffee berry sorting machine

Specific Objectives

1. To improve sorting mechanisms and algorithms to improve sorting accuracy.
2. To improve coffee berry ejection mechanisms
3. To implement co-processing architecture for faster processing between coffee berry detection, image processing and actuation.

1.4 Expected Outcomes

1. **Improved Sorting Accuracy** The improved sorting mechanisms and optimized algorithms are expected to significantly increase the accuracy of coffee berry classification. The system should reliably differentiate between ripe, unripe, defective, and foreign materials with minimal false positives and false negatives. This enhancement is expected to result in higher product quality, reduced manual intervention, and improved overall system efficiency.

2. **Enhanced Coffee Berry Ejection Performance** The improved ejection mechanism is expected to provide precise, timely, and consistent removal of identified defective or unwanted coffee berries. The system should demonstrate faster response times, reduced mis-ejections, and minimal mechanical wear. Additionally, the mechanism is expected to operate smoothly under continuous vibration conditions without causing berry damage or system instability.
3. **Faster Processing Through Co-Processing Architecture** The implementation of a co-processing architecture is expected to reduce system latency between detection, image processing, and actuation. By distributing tasks across multiple processing units, the system should achieve real-time or near real-time performance, increased throughput, and improved synchronization between subsystems. This is expected to enhance overall system responsiveness and allow for higher operational speeds without compromising sorting accuracy.

1.5 Justification

Coffee remains one of Kenya’s most valuable agricultural exports, and quality grading plays a major role in determining its economic value. Automated coffee berry sorting systems offer a practical solution for improving quality consistency and reducing labor dependency. However, many advanced commercial systems are prohibitively expensive and unsuitable for adoption by small-scale processors and cooperatives. Improving an existing image-based coffee berry sorting model provides a cost-effective pathway toward practical automation by building upon already validated concepts while addressing observed technical shortcomings. By focusing on real-time processing efficiency, motion-related detection challenges, and system responsiveness, this study aims to bridge the gap between prototype-level demonstrations and deployable solutions. The proposed improvements contribute to:

1. Enhanced accuracy and throughput in coffee berry sorting
2. Better utilization of affordable embedded vision technologies
3. Increased feasibility of locally developed agri-tech solutions
4. Practical learning and innovation in mechatronics system integration

Furthermore, incorporating real harvested coffee berries into testing and demonstration provides a realistic visual and operational representation of actual field conditions, strengthening the relevance and credibility of the system. The outcomes of this project are expected to support the development of scalable, affordable, and locally adaptable coffee processing technologies aligned with Kenya’s agricultural and technological development goals.

2 Literature Review

The global coffee industry plays a critical role in the economies of many developing countries, including Kenya, by providing livelihoods to millions of small-holder farmers. Coffee processing involves several stages, among which sorting of coffee berries is one of the most critical for determining final product quality and market value. The presence of unripe, overripe, damaged, or defective berries significantly affects cup quality, leading to reduced prices and rejection in premium markets. Traditionally, coffee berry sorting has relied on manual visual inspection, where workers separate berries based on perceived color, size, and visible defects. Although this approach has been practiced for decades, numerous studies have demonstrated that manual sorting is labor-intensive, inconsistent, slow, and highly subjective, particularly under fatigue and time pressure (Mugo et al., 2017; Silva Santos, 2019). Advances in machine vision,



Figure 2: Ripe coffee berries showing color variation from green to deep red

robotics, and embedded computing have enabled the development of automated coffee berry sorting systems capable of performing classification and separation tasks with higher accuracy, speed, and consistency. Automated systems typically integrate three core subsystems:

1. material handling mechanism (vibrating conveyor or chute)
2. A sensing and classification unit (camera and image processing algorithms)
3. separation mechanism (air jets, ejector plates, or diverters)

The 2019 automated coffee berry sorting machine project (reference project) represents a significant academic effort in this domain. The system employed a vibrating conveyor for controlled berry flow, a chute for alignment, image-based detection using a Raspberry Pi 3, and an air ejection mechanism for separation. While the system demonstrated the feasibility of automated sorting, several technical challenges were identified, particularly in real-time image processing and actuation timing, forming the foundation for the present study

2.1 Challenges of Traditional Sorting Methods

Manual sorting remains prevalent among small-scale farmers due to its low initial cost; however, extensive literature highlights its limitations.

1. **Labor Intensity and Fatigue:** Manual sorting requires a large workforce to maintain reasonable throughput. Studies show that prolonged visual inspection leads to fatigue, which directly reduces sorting accuracy and consistency [1]. Fatigued workers are more likely to misclassify berries, allowing defective berries to pass through or rejecting acceptable ones.
2. **Human Error and Subjectivity:** Manual sorting depends heavily on individual judgment, which varies between workers and across time [2]. Significant variability in sorting outcomes has been observed across different shifts, even when standardized grading guidelines were provided. This subjectivity results in inconsistent quality between batches.
3. **Limited Throughput:** Human sorting speed is inherently constrained. Manual sorting becomes a bottleneck during peak harvest seasons, limiting scalability and increasing processing delays [3]. This inefficiency directly affects farmers' ability to meet market demand.

These challenges have driven research into automated sorting systems capable of overcoming the limitations of manual methods.

2.2 Advantages of Automated Coffee Berry Sorting

Automated coffee berry sorting systems offer several key advantages over manual methods

1. **Increased Efficiency and Throughput:** Automated systems can process significantly larger volumes in shorter times compared to manual sorting. Modern optical sorting machines can handle hundreds of kilograms per hour, substantially reducing processing time during peak harvest.
2. **Improved Consistency and Accuracy:** Machine-based systems apply uniform classification criteria across all berries, eliminating subjective variation. Studies by Perez et al. (2020) demonstrate that automated systems maintain stable accuracy levels exceeding 92
3. **Reduced Labour Costs:** While requiring initial capital investment, automated sorting reduces ongoing labor requirements. Economic analysis by Mashauri et al. (2021) indicates that medium-scale processors can achieve cost recovery within 2–3 seasons through labor savings alone.
4. **Data Collection and Traceability:** Automated systems enable collection of processing statistics, including throughput rates, defect percentages, and batch-level quality metrics. This data supports quality management systems and provides traceability demanded by premium export markets.

2.3 Separation methods

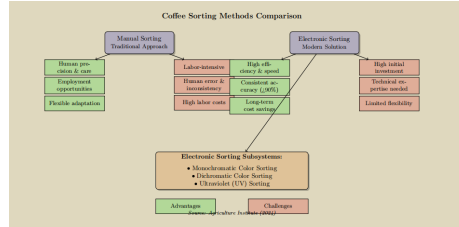


Figure 3: Comparison of manual and electronic coffee sorting methods (Source: Agriculture Institute, 2024)

2.3.1 Air Jet System

Air jet separation systems operate on a detect-decide-deflect principle. As berries pass through the imaging zone on a conveyor or chute, a high-speed camera (typically CCD or CMOS sensors) captures images in real time. Some advanced systems incorporate near infrared (NIR) sensors to detect internal defects not visible in the RGB spectrum. Image processing algorithms classify each berry, and defective berries trigger compressed air jets precisely timed to deflect them into reject chutes while acceptable berries continue along the primary path. Key advantages include:

1. Minimal physical contact reducing berry damage
2. Very high throughput capacity
3. Adaptability to various berry sizes without mechanical adjustment [5]

However, limitations include requirement for compressed air infrastructure, sensitivity to environmental conditions (wind, dust), and complexity in timing calibration.

2.3.2 Ejector plates

Ejector plate systems employ mechanical actuation to physically separate undesirable berries. When a defect is detected by upstream sensors, a solenoid or pneumatic actuator extends a small plate or paddle that pushes the target berry into a reject stream. This method ensures positive separation, meaning that once actuated, the berry is definitively moved to the reject pathway. Advantages include: high reliability in separation (no missed deflections due to air pressure variation), suitability for larger or heavier objects, and lower compressed air consumption. Limitations include: mechanical wear on actuators and contact surfaces, slower response times compared to air jets (20–50 ms), and potential for berry bruising at the contact point.

2.3.3 Diverter Systems

Diverter systems redirect entire groups or batches of berries rather than individual units. When a sensor detects a defective berry, a mechanical gate or diverter flap reroutes the material stream into a reject bin. This approach is typically used in lower-precision, bulk sorting scenarios where strict unit-level separation is not required. Advantages include: structural simplicity, low maintenance, and suitability for high volume, low-cost operations. Disadvantages include: lower precision (acceptable berries in the same batch may be rejected), inability to handle single-file streams, and inefficiency when defect rates are low.

2.4 Classification methods

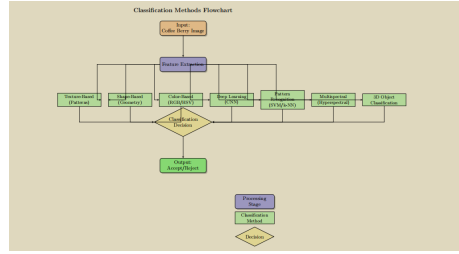


Figure 4: Overview of classification methods used in coffee berry sorting systems

2.4.1 Color-based classification

Color-based classification is the most widely used method in coffee berry sorting due to its direct correlation with maturity [6]. Ripe coffee berries exhibit deep red or purple hues, while unripe berries appear green or yellow. Systems employ RGB cameras to capture color information, which is then analyzed using color histograms, color space transformations (e.g., HSV, Lab), or simple thresholding. This method is cost-effective and computationally efficient, making it suitable for realtime processing on embedded platforms. However, it is highly dependent on uniform lighting conditions and can struggle with berries exhibiting intermediate ripeness states or surface discoloration from environmental factors.

2.4.2 Shape-Based Classification

Shape analysis assesses geometric attributes such as size, contour smoothness, aspect ratio, and circularity [7]. Machine vision systems extract berry contours from binary images and calculate shape descriptors to identify malformed, damaged, or insect-infested berries. This method complements color-based classification by detecting defects that do not manifest as color differences (e.g., physical

damage, irregular growth). However, shape analysis requires clear berry separation on the conveyor and consistent camera positioning to ensure measurement accuracy.

2.4.3 Texture-Based Classification

Texture-based methods analyze surface patterns using techniques such as Gray-Level CoOccurrence Matrix (GLCM), Local Binary Patterns (LBP), or Gabor filters. These approaches can detect surface defects like insect bites, mold growth, or drying-induced wrinkling. Texture analysis is particularly effective for identifying subtle defects not visible through color or shape alone. However, texture features are more computationally expensive to extract and may require higher-resolution imaging, which can limit real-time throughput on low-cost hardware.

2.4.4 Deep Learning-Based Classification

Deep learning approaches, particularly Convolutional Neural Networks (CNNs), have demonstrated superior classification accuracy by automatically learning discriminative features from labeled training data [8]. Architectures such as MobileNet [9], ResNet, and YOLO (You Only Look Once) [10] enable real-time object detection and classification. Deep learning models can handle complex, non-linear relationships between visual features and quality categories, achieving accuracies exceeding 95%. However, these models require substantial computational resources, large labeled datasets for training, and careful optimization to run efficiently on embedded platforms like the Raspberry Pi.

2.4.5 Pattern Recognition-Based Classification

Traditional pattern recognition methods employ supervised learning algorithms such as k-Nearest Neighbors (k-NN), Support Vector Machines (SVMs), or decision trees. These classifiers use hand-crafted features (color, shape, texture) as input and learn decision boundaries from training samples. Pattern recognition methods offer a balance between performance and computational efficiency. They are less demanding than deep learning approaches and more robust than simple thresholding, making them well-suited for embedded systems with limited processing power.

2.4.6 Multispectral and Hyperspectral Imaging-Based Classification

Multispectral and hyperspectral imaging capture information across multiple wavelengths, including visible and non-visible spectra (e.g., near-infrared, short-wave infrared). This allows detection of internal defects, moisture content variations, and chemical composition differences not visible to standard RGB cameras. These methods provide the highest classification accuracy and defect detection capability. However, the high cost of specialized cameras, the large data

volumes requiring processing, and the complexity of spectral analysis limit their adoption primarily to high-value, largescale industrial operations.

2.4.7 Three-Dimensional (3D) Object Classification

3D imaging systems use stereo vision, structured light, or time-of-flight sensors to capture depth information alongside 2D images. This enables measurement of berry volume, surface irregularities, and three-dimensional shape descriptors. 3D classification is valuable for detecting internal voids, volume-based sorting, and identifying deformities. However, 3D imaging systems are more expensive, require more complex calibration, and impose higher computational loads compared to 2D methods.

2.5 GAP ANALYSIS

Despite significant advances in automated coffee berry sorting, several critical gaps remain, particularly in the context of cost-effective, scalable solutions suitable for small-to-medium processors and university research environments.

Real-Time Processing and Throughput Limitations. The 5th-year project identified that the Raspberry Pi 3, while cost-effective, struggles to maintain real-time classification performance at higher throughput rates. The system experienced difficulties processing high frame rates, performing simultaneous object tracking and classification, and maintaining synchronization with real-time actuation signals. Although optimizations such as TensorFlow Lite deployment and multithreading were attempted, performance remained below the levels required for practical deployment. Current literature predominantly focuses on either high-performance industrial systems using GPUs or field-programmable gate arrays (FPGAs), which are cost-prohibitive, or low-throughput academic prototypes that do not address scalability. There is insufficient exploration of hybrid processing architectures where time-critical tasks are offloaded to co-processors (e.g., ESP32, STM32, or similar microcontrollers), allowing algorithm restructuring and task division between layers to achieve both cost efficiency and acceptable performance.

Motion Blur and Image Degradation at Higher Speeds

As conveyor speeds increase to meet higher throughput demands, image quality degrades due to motion blur, reducing classification accuracy. The 5th-year project explicitly identified motion blur as a major limitation, particularly at speeds exceeding certain thresholds. Existing solutions typically involve either reducing conveyor speed (sacrificing throughput) or increasing camera shutter speed and lighting intensity (increasing cost and power consumption). However, there is limited research on integrated approaches that combine algorithmic and mechanical mitigation strategies—such as adaptive frame capture triggering, region-of-interest (ROI) selective processing, and timing-aware rejection logic—that could address motion blur without disproportionately sacrificing speed or cost.

Cost-Performance Trade-Off in Hardware Selection There exists a notable gap in the literature regarding optimal hardware configurations for mid-scale sorting operations. High-performance systems using industrial-grade GPUs or dedicated vision processors offer excellent performance but are financially inaccessible to smallholder cooperatives and university prototypes. Conversely, ultra-low-cost platforms (e.g., Raspberry Pi alone) demonstrate limited processing speed and reduced scalability. Few studies systematically investigate cost-optimized improvements that maintain affordability while achieving meaningful performance gains. Strategies such as algorithm restructuring, selective processing pipelines, and intelligent task division between processing layers are underexplored. The current project aims to address this gap by evaluating hybrid processing architectures and system-level optimizations tailored to resource constrained environments.

3 Methodology

This section focuses on the modification and improvement of an existing image-based coffee berry sorting machine through the systematic redesign of its electrical and software subsystems. The mechanical subsystem was retained from the previous design, as it was structurally adequate for berry feeding and alignment. Therefore, the improvements concentrated on enhancing system processing capability, control coordination, and sorting performance.

The upgraded system utilizes a co-processing architecture incorporating a Raspberry Pi 4 and an Arduino Mega 2560 microcontroller to manage and coordinate key system components. The Raspberry Pi 4 performs image acquisition and high-level image processing tasks, while the Arduino Mega 2560 handles time-critical control operations and actuator interfacing. Together, they control four air ejectors, a vibrating conveyor mechanism, and the air compressor system.

Integration of a global shutter camera is important in order to capture clear images of coffee berries as they pass through the chute. The use of a global shutter camera minimizes motion blur, thereby improving image clarity during real-time operation. The captured images are processed using image detection algorithms to determine the ripeness and quality of each berry.

Based on the classification results, control signals are transmitted to the corresponding air ejectors, which are triggered at precise time intervals to separate the berries into predefined categories. The improved coordination between image processing and actuation enhances sorting accuracy, reduces response delays, and improves overall system throughput.

3.1 DESIGN

3.2 Mechanical design consideration

Sorting Mechanism Requirements .The sorting assembly must accurately differentiate berries based on ripeness and quality without causing physical damage. Mechanical alignment between the berry path and the detection zone is critical to ensure consistent image capture and precise actuation. The system must operate at a throughput rate that satisfies processing requirements while maintaining classification accuracy. Additionally, the mechanism should accommodate slight variations in berry size without requiring frequent recalibration.

Feeding System Performance. The feeding mechanism must regulate berry flow to prevent congestion, overlapping, or clustering at the inspection point. An uninterrupted and uniform feed ensures accurate image acquisition and minimizes misclassification. The design must also reduce impact forces and unnecessary bouncing to preserve berry integrity. Adequate feeder capacity is required to sustain continuous operation without repeated manual intervention. The feeding mechanism must regulate berry flow to prevent congestion, overlapping, or clustering at the inspection point. An uninterrupted and uniform feed ensures accurate image acquisition and minimizes misclassification. The design must also reduce impact forces and unnecessary bouncing to preserve berry integrity. Adequate feeder capacity is required to sustain continuous operation without repeated manual intervention. **Linear Vibrating Feeder Considerations.** The linear vibrating feeder must support stable and repeatable motion under continuous operation. Since coffee processing environments may be humid and dusty, the feeder components must withstand environmental exposure without corrosion or rapid wear. Smooth vibratory motion is essential to prevent erratic berry movement, which could negatively affect detection and ejection timing. The design must also allow easy access for cleaning and routine maintenance. The linear vibrating feeder must support stable and repeatable motion under continuous operation. Since coffee processing environments may be humid and dusty, the feeder components must withstand environmental exposure without corrosion or rapid wear. Smooth vibratory motion is essential to prevent erratic berry movement, which could negatively affect detection and ejection timing. The design must also allow easy access for cleaning and routine maintenance.

Material Selection Criteria. Materials selected for components in direct contact with coffee berries must be food-safe, non-reactive, and resistant to corrosion. Structural members must possess sufficient mechanical strength to withstand vibration, operational loads, and environmental exposure. Materials were selected based on durability, availability, cost-effectiveness, and suitability for agricultural processing environments. **Sensor and Vision Integration.** Mechanical positioning must ensure that the imaging system has a clear and stable field of view. The camera mounting structure must minimize vibration transmission from the feeder system to prevent image distortion. Proper alignment between the chute, camera, and ejectors is necessary to guarantee

accurate detection and actuation timing.

Actuation Mechanism Requirements. The air ejection system must deliver consistent and precisely directed air pulses to separate defective berries effectively. Actuators must support repeated high-frequency operation without significant performance degradation. Energy consumption and compressed air usage must be optimized to maintain system efficiency while ensuring reliable berry rejection.

3.3 Structural Subsystem

The structural subsystem consists of the following key components:

- **Linear Vibrating Feeder**

The linear vibrating feeder regulates the movement of coffee berries from the hopper toward the inspection zone. Controlled vibratory motion ensures gradual and uniform berry distribution, reducing clustering and improving detection consistency. The feeder design supports continuous operation while minimizing mechanical stress on the berries.

- **Hopper**

The hopper serves as the initial storage and loading unit for the coffee berries. Its geometry is designed to promote gravitational flow while minimizing bridging and clogging. The hopper ensures a steady supply of berries to the vibrating feeder during operation.

- **Chute**

The chute guides individual berries from the feeder to the inspection region. Its inclination angle and surface finish are designed to maintain controlled berry velocity and predictable trajectory, which is essential for accurate timing of detection and ejection.

- **Ejector Mounting Assembly**

The ejector mounting assembly secures and aligns the air nozzles relative to the berry trajectory. Proper positioning ensures that air pulses are delivered precisely at the required moment for effective separation. The mounting structure provides rigidity to prevent misalignment during operation.

- **Camera Mounting Structure**

The camera mounting structure supports the global shutter camera and associated lighting system. It is designed to isolate vibration and allow fine positional adjustments to achieve optimal focus and field coverage over the inspection zone.

- **Frame**

The frame acts as the primary support structure for all mechanical and electronic components. It maintains alignment between subsystems and absorbs operational loads generated by vibration and actuation. The frame is designed to ensure stability, structural integrity, and ease of assembly.

3.3.1 Linear Feeder Design

Linear feeders play a critical role in ensuring controlled and efficient transportation of coffee berries to the inspection and sorting stages. The selection of an appropriate feeder directly influences system throughput, sorting accuracy, and overall production efficiency.

Two commonly used feeder types in material handling systems are vibrating feeders and belt feeders. Vibrating feeders operate by generating controlled oscillatory motion that propels materials along a trough. This mechanism is particularly effective for handling bulk, small-sized, or granular materials, as it allows controlled flow regulation and easy adjustment of feed rate.

Belt feeders, on the other hand, transport materials using a continuous moving belt. They provide smoother and more uniform movement, making them suitable for handling larger or heavier products that require minimal agitation during transport.

For this project, a vibrating linear feeder was selected due to its ability to regulate berry flow, prevent clustering, and maintain controlled spacing between berries prior to image capture. This controlled motion is essential for accurate detection, tracking, and timely ejection within the sorting process.

Linear Feeder Material: The selection of a food-grade material for the vibrating feeder, suitable for handling coffee berries, involved evaluating common options such as PVC, polyurethane, aluminium, and stainless steel. Key considerations included durability, ease of cleaning, resistance to wear and tear, and the ability to withstand vibrations. Based on these criteria, stainless steel was determined to be the most suitable choice, offering superior rigidity, improved vibration transmission, and higher fatigue resistance compared to aluminium.

Table 1: Feeder Comparison

Feature	Vibrating Feeder	Belt Feeder
Flow Control	Consistent and adjustable flow	Consistent flow, but less adjustable
Throughput	High throughput with controlled feed rate	High throughput, but flow rate is less precise
Durability	Robust and durable construction	Durable, but belts can wear out and need replacement
Cost	Moderate initial cost, lower long-term costs	Lower initial cost, potentially higher long-term maintenance cost e.g belt replacements
Initial setup	Requires precise tuning of vibration parameters	Easier and quicker to set up initially

Table 2: Sample Table with 3 Columns and 6 Rows

Feature	Vibrating Feeder	Belt Feeder
Flow Control	Consistent and adjustable flow	Consistent flow, but less adjustable
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Initial setup	Requires precise tuning of vibration parameters	Easier and quicker to set up initially